

Chapter 5, Section 2

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1. If X is a birationally ruled surface, show that the curve C , such that X is birationally equivalent to $C \times \mathbb{P}^1$, is unique up to isomorphism.

Proof. Let C and C' be two nonsingular curves such that X is birational to $C \times \mathbb{P}^1$ and $C' \times \mathbb{P}^1$. Then, there exists rational maps $\pi, \pi' : X \dashrightarrow C, C'$ and sections $\sigma, \sigma' : C, C' \rightarrow X$ such that $\pi \circ \sigma$ and $\pi' \circ \sigma'$ extend to the identity morphism on C and C' , respectively. Then $\pi' \circ \sigma : C \rightarrow C'$ and $\pi \circ \sigma' : C' \rightarrow C$ are inverses to each other, so we have an isomorphism $C \cong C'$. $\square$

2. Let X be the ruled surface $\mathbb{P}(\mathcal{E})$ over a curve C . Show that $\mathcal{E}$ is decomposable if and only if there exist two sections C', C'' of X such that $C' \cap C'' = \emptyset$.

Proof. If $\mathcal{E}$ is decomposable, then $\mathcal{E} = \mathcal{O} \oplus \mathcal{L}$, where $\mathcal{L} = \mathcal{O}_C(\epsilon)$ for some divisor $\epsilon \in \text{Div } C$. By (2.8.1), (2.11.1), and (2.11.2), the two natural projections $\mathcal{E} \rightarrow \mathcal{L}$ and $\mathcal{E} \rightarrow \mathcal{O}_C$ correspond to sections $C' \sim C_0$ and $C'' \sim C_0 - \epsilon f$, respectively, such that $C' \cdot C'' = 0$. Hence, $C' \cap C'' = \emptyset$. Conversely, suppose there exist two sections C', C'' of X such that $C' \cap C'' = \emptyset$. The corresponding surjective morphisms $\mathcal{E} \rightarrow \mathcal{L}'$ and $\mathcal{E} \rightarrow \mathcal{L}''$ induce an $\mathcal{O}_C$ -module homomorphism $\mathcal{E} \rightarrow \mathcal{L}' \oplus \mathcal{L}''$. For every $P \in C$, the morphism of stalks $\mathcal{E}_P \rightarrow \mathcal{L}'_P \oplus \mathcal{L}''_P$ is an isomorphism because C' and C'' are disjoint, so $\mathcal{L}'_P \oplus \mathcal{L}''_P$ is generated by $\mathcal{E}_P$. Hence, $\mathcal{E} \rightarrow \mathcal{L}' \oplus \mathcal{L}''$ is an isomorphism. $\square$

3. (a) If $\mathcal{E}$ is a locally free sheaf of rank r on a nonsingular curve C , then there is a sequence

$$0 = \mathcal{E}_0 \subseteq \mathcal{E}_1 \subseteq \cdots \subseteq \mathcal{E}_r = \mathcal{E}$$

of subsheaves such that $\mathcal{E}_i / \mathcal{E}_{i-1}$ is an invertible sheaf for each $i = 1, \dots, r$. We say that $\mathcal{E}$ is a *successive extension* of invertible sheaves.

- (b) Show that this is false for varieties of dimension ≥ 2 . In particular, the sheaf of differentials $\Omega_{\mathbb{P}^2}$ on $\mathbb{P}^2$ is not an extension of invertible sheaves.

Proof.

- (a) We proceed by induction on r . The base case $r = 1$ is trivial, so assume the statement for any $r > 1$. We can twist $\mathcal{E}$ by a very ample sheaf $\mathcal{L}$ so that $\mathcal{E}' = \mathcal{E} \otimes \mathcal{L}$ is generated by global sections. If we prove the statement for $\mathcal{E}'$ so we have a filtration $(\mathcal{E}'_i)$, then tensoring by $\mathcal{L}^{-1}$ gives a filtration $(\mathcal{E}_i = \mathcal{E}'_i \otimes \mathcal{L}^{-1})$ of $\mathcal{E}$ such that

$$\mathcal{E}_i / \mathcal{E}_{i-1} \cong (\mathcal{E}'_i / \mathcal{E}'_{i-1}) \otimes \mathcal{L}^{-1},$$

which is an invertible sheaf for each $i = 1, \dots, r$. Hence, we can assume $\mathcal{E}$ is generated by global sections. The hypothesis of (II, Ex. 8.2) is satisfied for $\mathcal{E}$, so there exists a short exact

$$0 \longrightarrow \mathcal{O}_C \longrightarrow \mathcal{E} \longrightarrow \mathcal{F} \longrightarrow 0,$$

where $\mathcal{F}$ is also globally generated locally free of rank $r - 1$. By the inductive hypothesis, this means we can find a filtration $(\mathcal{F}_i)$ of $\mathcal{F}$ such that each successive quotient is an invertible sheaf. Consider the preimage $(\mathcal{E}_j)$ of $(\mathcal{F}_i)$, where we relabel $j = i + 1$. Each $\mathcal{E}_j$ is also locally free, $\mathcal{E}_1 = \text{im}(\mathcal{O}_C)$ by exactness, and there is a canonical isomorphism $\mathcal{E}_j / \mathcal{E}_{j-1} \cong \mathcal{F}_{j-1} / \mathcal{F}_{j-2}$. Extending $(\mathcal{E}_j)$ with $\mathcal{E}_0 = 0$ gives a successive extension of sheaves.

- (b) Suppose we have an exact sequence

$$0 \longrightarrow \mathcal{L} \longrightarrow \Omega \longrightarrow \mathcal{M} \longrightarrow 0,$$

where $\mathcal{L}$ and $\mathcal{M}$ are invertible sheaves on $\mathbb{P}^2$. Then $H^1(\mathcal{L}), H^1(\mathcal{M}) = 0$, so by the long exact sequence in cohomology, we have $H^1(\Omega) = 0$. But by (III, Ex. 7.2), $H^1(\Omega) = k$, a contradiction. $\square$

4. Let C be a curve of genus g , and let X be the ruled surface $C \times \mathbb{P}^1$. We consider the question, for what integers $s \in \mathbb{Z}$ does there exist a section D of X with $D^2 = s$? First show that s is always an even integer, say $s = 2r$.

- (a) Show that $r = 0$ and $r \geq g + 1$ are always possible.
- (b) If $g = 3$, show that $r = 1$ is not possible, and just one of the two values $r = 2, 3$ is possible, depending on whether C is hyperelliptic or not.

Proof. Fix a section $C_0 = C \times P_0 \subset X$, where P_0 is a point of $\mathbb{P}^1$. The ruled surface $X = C \times \mathbb{P}^1$ corresponds to the projective bundle of $\mathcal{E} = \mathcal{O}_C \oplus \mathcal{O}_C$. The invariant of X is $e = 0$, which means $C_0^2 = -e = 0$. By (2.9), if D is any section of X corresponding to a surjection $\mathcal{E} \rightarrow \mathcal{O}_C(\mathfrak{d})$ for some divisor $\mathfrak{d}$ of C with $r = \deg \mathfrak{d}$, then $r = C_0 \cdot D$ and $D \sim C_0 + \mathfrak{d}f$. Hence, $s = 2r$.

- (a) There is a one-to-one correspondence between sections D of X and finite separable morphisms $f : C \rightarrow \mathbb{P}^1$. In particular, a section D is the graph of a morphism $f : C \rightarrow \mathbb{P}^1$, and its self-intersection is $D^2 = 2 \cdot \deg f$. By (IV, Ex. 1.6), $g + 1$ is an upperbound on the minimum of $\deg f$. Hence, $r = 0$ and $r \geq g + 1$ is always possible.
- (b) If $g > 0$, then there is no morphism $C \rightarrow \mathbb{P}^1$ of degree 1. If C is a hyperelliptic curve, then it has a g_2^1 , so $r = 2, 3$ is possible. Otherwise, nonhyperelliptic curves of genus 3 always have a g_3^1 (IV, 5.5.2), so $r = 3$ is always possible.

□

5. *Values of e .* Let C be a curve of genus $g \geq 1$.

- (a) Show that for each $0 \leq e \leq 2g - 2$ there is a ruled surface X over C with invariant e , corresponding to an indecomposable $\mathcal{E}$.
- (b) Let $e < 0$, let D be any divisor of degree $d = -e$, and let $\xi \in H^1(X, \mathcal{O}(-D))$ be a nonzero element defining an extension

$$0 \longrightarrow \mathcal{O} \longrightarrow \mathcal{E} \longrightarrow \mathcal{O}(D) \longrightarrow 0.$$

Let $H \subseteq |D + K|$ be the sublinear system of codimension 1 defined by $\ker \xi$, where ξ is considered as a linear functional on $H^0(\mathcal{O}(D + K))$. For any effective divisor E of degree $d - 1$, let $L_E \subseteq |D + K|$ be the sublinear system $|D + K - E| + E$. Show that $\mathcal{E}$ is normalized if and only if for each E as above, $L_E \not\subseteq H$.

- (c) Now show that if $-g \leq e < 0$, there exists a ruled surface X over C with invariant e .
- (d) For $g = 2$, show that $e \geq -2$ is also necessary for the existence of X .

Proof.

- (a) Let $\mathfrak{e}$ be a divisor of degree $-e$ on C , and let $\mathfrak{k}$ is a canonical divisor of C . We need to first construct an indecomposable extension $\mathcal{E}$ of $\mathcal{O}(\mathfrak{e})$ by $\mathcal{O}$. By (III, Ex. 6.1), an extension of vector bundles

$$0 \longrightarrow \mathcal{O} \longrightarrow \mathcal{E} \longrightarrow \mathcal{O}(\mathfrak{e}) \longrightarrow 0$$

corresponds to an element in $\text{Ext}^1(\mathcal{O}(\mathfrak{e}), \mathcal{O})$. In this case, $\text{Ext}^1(\mathcal{O}(\mathfrak{e}), \mathcal{O}) \cong H^1(\mathcal{O}(-\mathfrak{e}))$ (III, 6.3) and (III, 6.7). This is dual to $H^0(\mathcal{O}(\mathfrak{k} + \mathfrak{e}))$, so it is enough to find $\mathfrak{e}$ such that $H^0(\mathcal{O}(\mathfrak{k} + \mathfrak{e})) \neq 0$. The image of the canonical morphism $C \rightarrow |\mathfrak{k}| \cong \mathbb{P}^{g-1}$ is either a rational curve of degree $g - 1$, or an embedding of degree $2g - 2$, depending on whether or not C is hyperelliptic. If $-\mathfrak{e}$ is effective, then $H^0(\mathcal{O}(\mathfrak{k} + \mathfrak{e}))$ is the space of all hyperplane sections in $\mathbb{P}^{g-1}$ with support containing $\mathfrak{e}$. In any case, a hyperplane section of C in $\mathbb{P}^{g-1}$ consists of $2g - 2$ points counted with multiplicity. Hence, for each $0 \leq e \leq 2g - 2$, there is a divisor $\mathfrak{e}$ of degree $-e$ such that $H^0(\mathcal{O}(\mathfrak{k} + \mathfrak{e})) \neq 0$, so there exists indecomposable extensions of $\mathcal{O}(\mathfrak{e})$ by $\mathcal{O}$.

Now, let $\mathcal{E}$ be an extension of $\mathcal{O}(\mathfrak{e})$ defined by a nonzero element in $\text{Ext}^1(\mathcal{O}(\mathfrak{e}), \mathcal{O})$. It remains to show $\mathcal{E}$ is normalized and indecomposable. Clearly $H^0(\mathcal{E}) \neq 0$ by construction. If $\mathcal{M}$ is any invertible sheaf, then we have an exact sequence

$$0 \longrightarrow \mathcal{M} \longrightarrow \mathcal{E} \otimes \mathcal{M} \longrightarrow \mathcal{O}(\mathfrak{e}) \otimes \mathcal{M} \longrightarrow 0.$$

If $\deg \mathcal{M} < 0$, then we have $H^0(\mathcal{M}) = 0$ and $H^0(\mathcal{O}(\mathfrak{e}) \otimes \mathcal{M}) = 0$, and therefore also $H^0(\mathcal{E} \otimes \mathcal{M}) = 0$. Furthermore, if $\mathcal{E}$ is decomposable, then it would be $\mathcal{O} \oplus \mathcal{L}$ for some $\mathcal{L}$, by (2.12). But $\bigwedge^2 \mathcal{E} \cong \mathcal{O}_C(\mathfrak{e})$, so $\mathcal{L} \cong \mathcal{O}(\mathfrak{e})$, so in fact this extension would have to split, which it doesn't. Hence, for each $0 \leq e \leq 2g - 2$, there is a ruled surface X over C with invariant e , corresponding to an indecomposable $\mathcal{E}$.

- (b) Let $\xi \in H^1(\mathcal{O}(-D))$ be the (non-zero) element defining the extension $\mathcal{E}$. If E is any effective divisor of degree $d - 1$ on C , then we have a commutative diagram

$$\begin{array}{ccc} H^0(\mathcal{O}) & \xrightarrow{\delta} & H^1(\mathcal{O}(-D)) \\ \alpha \downarrow & & \beta \downarrow \\ H^0(\mathcal{O}(E)) & \xrightarrow{\gamma} & H^1(\mathcal{O}(E - D)) \end{array}$$

where $\delta(1) = \xi$, $\alpha(1) = t$, a non-zero section defining E , and β is induced from the map $\mathcal{O} \rightarrow \mathcal{O}(E)$ corresponding to t . On the other hand, β is dual to the map

$$\beta^\vee : H^0(\mathcal{O}(D + K - E)) \rightarrow H^0(\mathcal{O}(D + K))$$

also induced by t . The subspace of $H^0(\mathcal{O}(D + K))$ corresponding to L_E is precisely the image of $\beta^\vee$, so $\beta(\xi) = 0$ if and only if $L_E \subseteq H$.

If $\mathcal{E}$ is normalized, then γ is injective for any effective divisor E of degree $d - 1$. By the commutative diagram, we must have $\beta(\xi) \neq 0$. Hence, $L_E \not\subseteq H$.

Conversely, if $\mathcal{E}$ is not normalized, then $H^0(\mathcal{E} \otimes \mathcal{M}) \neq 0$ for some invertible sheaf $\mathcal{M}$ with $\deg \mathcal{M} < 0$, and let $m = -\deg \mathcal{M}$. We have the exact sequence

$$0 \longrightarrow \mathcal{M} \longrightarrow \mathcal{E} \otimes \mathcal{M} \longrightarrow \mathcal{O}(D) \otimes \mathcal{M} \longrightarrow 0.$$

If $m \geq d$, then we have $H^0(\mathcal{E} \otimes \mathcal{M}) = 0$ since $\deg \mathcal{O}(D) \otimes \mathcal{M} < 0$, and if $m = d$, then either $H^0(\mathcal{O}(D) \otimes \mathcal{M}) = 0$, or $\mathcal{M} = \mathcal{O}(-D)$, whence the coboundary map δ is injective, hence $H^0(\mathcal{E} \otimes \mathcal{M}) = 0$. So we can assume $0 < m < d$. The long exact sequence of cohomology gives

$$0 \longrightarrow H^0(\mathcal{E} \otimes \mathcal{M}) \longrightarrow H^0(\mathcal{O}(D) \otimes \mathcal{M}) \longrightarrow H^1(\mathcal{M}),$$

where $H^0(\mathcal{M}) = 0$. The middle term is non-zero, so if $\mathcal{M} \cong \mathcal{O}(-M)$ with $m = \deg M$, then $D - M \sim E$ for some effective divisor E of degree $d - m$. For any effective divisor $\tilde{E}$ of degree $d - 1$ with $\tilde{E} - E \geq 0$, we have a commutative diagram,

$$\begin{array}{ccc} H^0(\mathcal{O}) & \xrightarrow{\delta} & H^1(\mathcal{O}(-D)) \\ \alpha \downarrow & & \beta \downarrow \\ H^0(\mathcal{O}(E)) & \xrightarrow{\gamma} & H^1(\mathcal{O}(E - D)) \\ \phi \downarrow & & \psi \downarrow \\ H^0(\mathcal{O}(\tilde{E})) & \xrightarrow{\eta} & H^1(\mathcal{O}(\tilde{E} - D)) \end{array} \quad \begin{array}{c} \tilde{\alpha} \curvearrowright \\ \tilde{\beta} \curvearrowleft \end{array}$$

where $\delta(1) = \xi$, α, β is induced by a section $t \in H^0(\mathcal{O}(E))$ defining E , and ϕ, ψ is induced by a section s defining $\tilde{E} - E$.

Now, if $\beta(\xi) = 0$, then we are done. Otherwise, dualizing the column on the right of the commutative diagram gives

$$\begin{array}{ccc} H^0(\mathcal{O}(D + K - \tilde{E})) & & \\ \downarrow \psi^\vee & \searrow \tilde{\beta}^\vee & \\ H^0(\mathcal{O}(D + K - E)) & \xrightarrow{\beta^\vee} & H^0(\mathcal{O}(D + K)) \end{array}.$$

Thus, for every $\tilde{E}$, we have $L_{\tilde{E}} \subseteq L_E$. By the assumption $\beta(\xi) \neq 0$, we have $L_E \not\subseteq H$, so $\text{im}(\beta^\vee) \cap \ker(\xi)$ is a nonempty proper subspace of $\text{im}(\beta^\vee)$. By the Riemann-Roch theorem, we have

$$\dim H^0(\mathcal{O}(D + K - \tilde{E})) = g, \quad \dim H^0(\mathcal{O}(D + K - E)) = g + m - 1.$$

The space of possible $\tilde{E}$ has dimension $m - 1$, so the image of $\psi^\vee$ as $\tilde{E}$ varies is dense inside $H^0(\mathcal{O}(D + K - E))$. Consequently, we can find at least one $\tilde{E}$ such that $\text{im}(\tilde{\beta}^\vee) \not\subseteq \ker(\xi)$, which is what we wanted to show.

- (c) Let $\mathbb{P}^N = |K + D|$ with $N = g + d - 2$. For any effective divisor E of degree $d - 1$ on C , let us consider the set

$$B_E = \{\text{hyperplanes } H \subset \mathbb{P}^N \mid L_E \subseteq H\} \subset |H|,$$

where $|H|$ is the complete linear system of hyperplanes in $\mathbb{P}^N$, i.e., $|H| = |K + D|$. A hyperplane in $\mathbb{P}^N$ is determined by a nonzero global section $f \in V = H^0(\mathbb{P}^N, \mathcal{O}_{\mathbb{P}^N}(1))$. Viewing L_E as a linear subspace of $\mathbb{P}^N$, let $W_E = H^0(L_E, \mathcal{O}_{L_E}(1))$. The space of effective divisors of degree $d - 1$ is parameterized by the closed points of the symmetric product $C^{(d-1)}$, which has dimension $d - 1$. Consider the restriction map

$$\varphi_E : V \rightarrow W_E.$$

It is clearly surjective, and the hyperplanes $H \in B_E$ correspond exactly to those $f \in \ker \varphi_E$. We have $\dim V = g + d - 1$ and $\dim W_E = g$, so $\dim B_E = \dim(\ker \varphi_E) - 1 = d - 2$.

Now, consider the subset $B \subset C^{(d-1)} \times |H|$ consisting of all pairs $\langle E, H \rangle$ such that $E \in C^{(d-1)}$ is a closed point and $H \in B_E$. Clearly B is the set of closed points of a $C^{(d-1)} \times |H|$, which we denote also by B , and which we give a reduced induced scheme structure. The first projection $p_1 : B \rightarrow C^{(d-1)}$ is surjective, with fibre a projective space of dimension $d - 2$. So B has dimension $(d - 1) + (d - 2) = 2d - 3$, which is strictly smaller than $\dim |H| = g + d - 2$. If $H \in |H| - p_2(B)$, then $H \not\supseteq L_E$ for every $E \in C^{(d-1)}$, so the extension of $\mathcal{O}(D)$ by $\mathcal{O}$ given by the element in $H^0(\mathcal{O}(K + D)) \cong H^1(\mathcal{O}(-D))$ corresponding to H satisfies the requirement of (b). Hence, if $-g \leq e < 0$, then there exists a ruled surface X over C with invariant e .

- (d) Let C be a curve of genus $g = 2$, let $\mathcal{E}$ be a normalized locally free sheaf of rank 2 on C with invariant e , let $d = -e$, and let $X = \mathbb{P}(\mathcal{E})$ be the ruled surface associated to $\mathcal{E}$. As in the proof of (2.15), we want to show $e = -3, -4$ does not occur.

If it did, we would have a normalized bundle $\mathcal{E}$ with an exact sequence

$$0 \longrightarrow \mathcal{O} \longrightarrow \mathcal{E} \longrightarrow \mathcal{O}(D) \longrightarrow 0,$$

where D is an effective divisor of degree $d = -e$. Let $\xi \in H^1(\mathcal{O}(-D))$ be the element defining the extension $\mathcal{E}$. If E is any effective divisor, then applying $\text{Hom}(\mathcal{O}(D - E), -)$ and taking the long exact sequence in cohomology gives

$$\text{Hom}(\mathcal{O}(D - E), \mathcal{E}) \longrightarrow \text{Hom}(\mathcal{O}(D - E), \mathcal{O}(D)) \xrightarrow{\delta_E} \text{Ext}^1(\mathcal{O}(D - E), \mathcal{O}),$$

so by exactness, the inclusion $i : \mathcal{O}(D - E) \rightarrow \mathcal{O}(D)$ factors through $\mathcal{E} \rightarrow \mathcal{O}(D)$ if and only if $\delta_E(i) = 0$.

By naturality, we have a commutative diagram

$$\begin{array}{ccc} \text{Hom}(\mathcal{O}(D), \mathcal{O}(D)) & \xrightarrow{\delta} & \text{Ext}^1(\mathcal{O}(D), \mathcal{O}) \\ \downarrow i^* & & \downarrow i^* \\ \text{Hom}(\mathcal{O}(D - E), \mathcal{O}(D)) & \xrightarrow{\delta_E} & \text{Ext}^1(\mathcal{O}(D - E), \mathcal{O}) \end{array},$$

where $\delta(1) = \xi$ and $i^*(1) = i$, so $\delta_E(i) = 0$ if and only if $i^*(\xi) = 0$. Now, if there exists E of degree $< d$ such that $i : \mathcal{O}(D - E) \rightarrow \mathcal{E}$ factors through $\mathcal{E} \rightarrow \mathcal{O}(D)$, then $\mathcal{O}(D - E) \rightarrow \mathcal{E}$ must also be injective, which is a contradiction of $\mathcal{E}$ being normalized, so it is enough to show there exists E such that $i^*(\xi) = 0$.

The discussion above has the following geometric interpretation. By Serre duality, there is a natural identification

$$|D + K| = \mathbb{P}(\text{Ext}^1(\mathcal{O}(D), \mathcal{O})) = \mathbb{P}(H^0(\mathcal{O}(D + K))^\vee),$$

so we can view the linear span of ξ in $\text{Ext}^1(\mathcal{O}(D), \mathcal{O})$ as a point in $|D + K|$, which we will also denote by ξ . It follows that $i^*(\xi) = 0$ if and only if ξ is contained in every hyperplane in $|D + K|$ which also contains E .

$d = 3$ In either case, $D + K$ is a very ample divisor on C (IV, Ex. 3.1), so consider the embedding of C in $|D + K| \cong \mathbb{P}^3$. The secant variety of C fills up the entire space (IV, §3), so there is a secant line $L \subseteq |D + K|$ which contains ξ . A line is uniquely determined by two points, so we can take E to be any effective divisor of degree 2 with support contained in $C \cap L$. Hence, any hyperplane $H \subseteq |D + K|$ containing E also contains L , so ξ must be a point in H as well.

$d = 4$ Again, consider C in its embedding in $|D + K| \cong \mathbb{P}^4$. A plane is uniquely determined by three points, and the union of all planes going through at least three points of C fills up all of $|D + K|$. This means we can find a plane P in $|D + K|$ which contains ξ and at least three points of C . So we can take E to be any effective divisor of degree 3 with support contained in $C \cap P$. Hence, any hyperplane $H \subseteq |D + K|$ containing E also contains P , so ξ must be in H as well.

□

6. Show that every locally free sheaf of finite rank on $\mathbb{P}^1$ is isomorphic to a direct sum of invertible sheaves.

Proof. We call any locally free sheaf $\mathcal{E}$ on a curve C *normalized* if $H^0(C, \mathcal{E}) \neq 0$ and $H^0(C, \mathcal{E} \otimes \mathcal{L}) = 0$ for all invertible sheaves $\mathcal{L}$ on C with $\deg \mathcal{L} < 0$. With this definition, the statement of (2.8) can be made sense for $\mathcal{E}$ of arbitrary rank:

Lemma 0.1. *For every locally free sheaf $\mathcal{E}$ of finite rank on a curve C , there exists an invertible sheaf $\mathcal{L}$ on C such that $\mathcal{E} \otimes \mathcal{L}$ is normalized.*

Going back to the original statement, we can assume $\mathcal{E}$ is normalized. We proceed by induction on the rank r of $\mathcal{E}$. The base case $r = 1$ is trivial, so assume the statement holds for any $< r$. Pick a subinvertible sheaf $0 \rightarrow \mathcal{O} \rightarrow \mathcal{E}$ of maximal degree, and consider the short exact sequence

$$0 \longrightarrow \mathcal{O} \longrightarrow \mathcal{E} \longrightarrow \mathcal{F} \longrightarrow 0.$$

The quotient sheaf $\mathcal{F} = \mathcal{E}/\mathcal{O}$ is locally free of rank $r - 1$ because of the choice of $0 \rightarrow \mathcal{O} \rightarrow \mathcal{E}$ (2.8), so by the inductive hypothesis, $\mathcal{F}$ is decomposable, say $\mathcal{F} = \bigoplus \mathcal{O}(b_i)$ for some $b_i \in \mathbb{Z}$. Tensoring by $\mathcal{O}(-1)$ and taking cohomology gives

$$0 \longrightarrow H^0(\mathcal{O}(-1)) \longrightarrow H^0(\mathcal{E}(-1)) \longrightarrow H^0(\bigoplus \mathcal{O}(b_i - 1)) \longrightarrow 0.$$

Since $\mathcal{E}$ is normalized, the middle vector space is zero, so by exactness we have $b_i \leq 0$. On the other hand, the space of extensions of $\mathcal{F}$ by $\mathcal{O}$ can be computed by Serre duality, which gives

$$\text{Ext}^1(\mathcal{F}, \mathcal{O}) \cong H^0(\bigoplus \mathcal{O}(b_i - 1)) = 0,$$

so any extension of $\mathcal{F}$ by $\mathcal{O}$ is trivial. Hence, the short exact sequence splits, and there is an isomorphism $\mathcal{E} \cong \mathcal{O} \oplus \mathcal{F}$. $\square$

7. On the elliptic ruled surface X of (2.11.6), show that the sections C_0 with $C_0^2 = 1$ form a one-dimensional algebraic family, parametrized by the points of the base curve C , and that no two are linearly equivalent.

Proof. Combine (III, Ex. 11.5), (2.9), and the fact that any divisor of positive degree on an elliptic curve is linearly equivalent to an effective divisor. $\square$

8. A locally free sheaf $\mathcal{E}$ on a curve C is said to be *stable* if for every quotient locally free sheaf $\mathcal{E} \rightarrow \mathcal{F} \rightarrow 0$, $\mathcal{F} \neq \mathcal{E}$, $\mathcal{F} \neq 0$, we have

$$(\deg \mathcal{F})/\text{rank } \mathcal{F} > (\deg \mathcal{E})/\text{rank } \mathcal{E}$$

Replacing $>$ by $\geq$ defines *semistable*.

- (a) A decomposable $\mathcal{E}$ is never stable.
- (b) If $\mathcal{E}$ has rank 2 and is normalized, then $\mathcal{E}$ is stable (respectively, semistable) if and only if $\deg \mathcal{E} > 0$ (respectively, ≥ 0).
- (c) Show that the indecomposable locally free sheaves $\mathcal{E}$ of rank 2 that are not semistable are classified, up to isomorphism, by giving (1) an integer $0 < e \leq 2g - 2$, (2) an element $\mathcal{L} \in \text{Pic } C$ of degree $-e$, and (3) a nonzero $\xi \in H^1(C, \mathcal{L}^{-1})$, determined up to a nonzero scalar multiple.

Proof.

- (a) Denote $\mu(\mathcal{E}) = (\deg \mathcal{E})/\text{rank } \mathcal{E}$ the slope of $\mathcal{E}$, and let $\mathcal{F} \subseteq \mathcal{E}$ consist of the summands of $\mathcal{E}$ that has degree $\leq \mu(\mathcal{E})$, which there must be at least one. There is a surjective morphism of locally free sheaves $\mathcal{E} \rightarrow \mathcal{F} \rightarrow 0$, and we have $\mu(\mathcal{F}) \leq \mu(\mathcal{E})$. Hence, $\mathcal{E}$ is not stable.
- (b) If $\mathcal{E}$ is normal with invariant $e = -\deg \mathcal{E}$, then we have a quotient locally sheaf $\mathcal{E} \rightarrow \mathcal{L} \rightarrow 0$, where $\mathcal{L}$ is an invertible sheaf of degree $-e$. If $\mathcal{E}$ is stable (respectively, semistable), then

$$-e = \mu(\mathcal{L}) > \mu(\mathcal{E}) = -e/2,$$

(respectively, $\mu(\mathcal{L}) \geq \mu(\mathcal{E})$), which implies $\deg \mathcal{E} = -e > 0$ (respectively, ≥ 0). Conversely, suppose $\deg \mathcal{E} > 0$ (respectively, ≥ 0), and let $\mathcal{E} \rightarrow \mathcal{F} \rightarrow 0$ be a quotient locally free sheaf. If $\mathcal{F}$ has rank 2, then $\mathcal{E} \rightarrow \mathcal{F} \rightarrow 0$ is an isomorphism (II, Ex. 7.1). Otherwise, $\mathcal{F}$ is an invertible sheaf, and because of the normalization of $\mathcal{E}$, we must have $\deg \mathcal{F} \geq \deg \mathcal{E}$. Hence, $\mu(\mathcal{F}) = \deg \mathcal{F} \geq \deg \mathcal{E} > \mu(\mathcal{E})$ (respectively, $\geq \mu(\mathcal{E})$) if $\deg \mathcal{E} > 0$ (respectively, $\deg \mathcal{E} \geq 0$).

- (c) There is a small typo in the problem statement. The classification should concern *normalized* indecomposable locally free sheaves of rank 2 that are not semistable. Combine (III, Ex. 6.1), (III, Ex. 12.5), (b), and the fact that the extension $0 \rightarrow \mathcal{O} \rightarrow \mathcal{E} \rightarrow \mathcal{L} \rightarrow 0$ defined by $\xi \in H^1(C, \mathcal{L}^{-1})$ is normalized if $e > 0$.

□

9. Let Y be a nonsingular curve on a quadric cone X_0 in $\mathbb{P}^3$. Show that either Y is a complete intersection of X_0 with a surface of degree $a \geq 1$, in which case $\deg Y = 2a$, $g(Y) = (a-1)^2$, or $\deg Y$ is odd, say $2a+1$, and $g(Y) = a^2 - a$.

Proof. A general hyperplane section of X_0 in $\mathbb{P}^3$ is a conic curve C , and X_0 is a cone over C in $\mathbb{P}^3$ with vertex P_0 . If we blow up the point P_0 as in (2.11.4), then we obtain the rational ruled surface $\mathbb{F}$ with invariant $e = 2$, such that if $\pi : \mathbb{F} \rightarrow X_0$ is the projection map, then $\pi^{-1}(P_0)$ is a section C_0 with $C_0^2 = -2$. In other words, $\mathbb{F}$ is isomorphic to the projective bundle of $\mathcal{E} = \mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(-2)$ over $\mathbb{P}^1$. The hyperplane sections of $\pi : \mathbb{F} \rightarrow X_0 \hookrightarrow \mathbb{P}^3$ correspond to the linear system $|\mathcal{O}_{\mathbb{F}}(1)| = |C_0 + 2f|$, and by (2.11), $K = -2C_0 - 4f$ is a canonical divisor of $\mathbb{F}$. If Y is a complete intersection of X_0 with a surface of degree $a \geq 1$, then $Y \sim a(C_0 + 2f)$, so $\deg Y = Y \cdot (C_0 + 2f) = 2a$, and the adjunction formula gives

$$g = \frac{1}{2}Y \cdot (Y + K) + 1 = (a-1)^2.$$

Conversely, if Y has even degree, say $\deg Y = 2a$, then $Y \sim a(C_0 + 2f) - bC_0$ with $0 \leq b < a$ (1.3) (2.20). Let $\mathcal{I} = \mathcal{O}_{\mathbb{F}}(-Y)$ be the ideal sheaf of Y as a closed subscheme of $\mathbb{F}$. Tensoring by $\mathcal{O}_{\mathbb{F}}(a)$, taking global sections, and applying (2.4) gives

$$H^0(\mathbb{F}, \mathcal{I}(a)) \cong H^0(\mathbb{P}^1, S^b(\mathcal{E})),$$

where we have

$$S^b(\mathcal{E}) \cong \bigoplus_{i=0}^b \mathcal{O}_{\mathbb{P}^1}(-2i).$$

It follows that there exists a section in $|\mathcal{O}_{\mathbb{F}}(a)|$ containing Y , and since $H^0(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(a)) \rightarrow H^0(\mathbb{F}, \mathcal{O}_{\mathbb{F}}(a)) \rightarrow 0$ is surjective, we can find a homogenous polynomial of degree a in $\mathbb{P}^3$ containing Y . So by reasons of degree, Y is the complete intersection of X_0 with a surface of degree a .

If Y has odd degree, say $\deg Y = 2a+1$, then for the same reason in the case $\deg Y$ is even, Y is contained in a surface of degree $a+1$, so let Z be any such surface in $\mathbb{P}^3$. By (I, 7.7) $X_0 \cap Z$ has two irreducible components each of multiplicity one, consisting of Y and a line L in $\mathbb{P}^3$. The latter must be a member of the one-dimensional family of lines in X_0 which go through the vertex P_0 , and the inverse image $L' = L \times_{X_0} X$ of any of these lines is linearly equivalent to $C_0 + f$, which means $Y \sim aC_0 + (2a+1)f$. Now, we can compute the genus of Y using the adjunction formula of (1.5)

$$g = \frac{1}{2}Y \cdot (Y + K) + 1 = a^2 - a.$$

□

10. For any $n > e \geq 0$, let X be the rational scroll of degree $d = 2n - e$ in $\mathbb{P}^{d+1}$ given by (2.19). If $n \geq 2e - 2$, show that X contains a nonsingular curve Y of genus $g = d + 2$ which is a canonical curve in this embedding. Conclude that for every $g \geq 4$, there exists a nonhyperelliptic curve which has a g_3^1 .

Proof.

□

11. Let X be a ruled surface over the curve C , defined by a normalized bundle $\mathcal{E}$, and let $\mathfrak{e}$ be the divisor on C for which $\mathcal{O}_C(\mathfrak{e}) \cong \bigwedge^2 \mathcal{E}$ (2.8.1). Let $\mathfrak{b}$ be any divisor on C .
- (a) If $|\mathfrak{b}|$ and $|\mathfrak{b} + \mathfrak{e}|$ have no base points, and if $\mathfrak{b}$ is nonspecial, then there is a section $D \sim C_0 + \mathfrak{b}f$, and $|D|$ has no points.
 - (b) If $\mathfrak{b}$ and $\mathfrak{b} + \mathfrak{e}$ are very ample on C , and for every point $P \in C$, we have $\mathfrak{b} - P$ and $\mathfrak{b} + \mathfrak{e} - P$ nonspecial, then $C_0 + \mathfrak{b}f$ is very ample.
12. Let X be a ruled surface with invariant e over an elliptic curve C , and let $\mathfrak{b}$ be a divisor on C .
- (a) If $\deg \mathfrak{b} \geq \mathfrak{e} + 2$, then there is a section $D \sim C_0 + \mathfrak{b}f$ such that $|D|$ has no base points.
 - (b) The linear system $|C_0 + \mathfrak{b}f|$ is very ample if and only if $\deg \mathfrak{b} \geq e + 3$.
13. For every $e \geq -1$ and $n \geq e + 3$, there is an elliptic scroll of degree $d = 2n - e$ in $\mathbb{P}^{d-1}$. In particular, there is an elliptic scroll of degree 5 in $\mathbb{P}^4$.
14. Let X be a ruled surface over a curve of genus g , with invariant $e < 0$, and assume that $\text{char } k = p > 0$ and $g \geq 2$.
- (a) If $Y \equiv aC_0 + bf$ is an irreducible curve $\neq C_0.f$, then either $a = 1, b \geq 0$, or $2 \leq a \leq p - 1, b \geq \frac{1}{2}ae$, or $a \geq p, b \geq \frac{1}{2}ae + 1 - g$.
 - (b) If $a > 0$ and $b > a(\frac{1}{2}e + (1/p)(g - 1))$, then any divisor $D \equiv aC_0 + bf$ is ample. On the other hand, if D is ample, then $a > 0$ and $b > \frac{1}{2}ae$.
15. *Funny behaviour in characteristic p .* Let C be the plane curve $x^3y + y^3z + z^3x = 0$ over a field k of characteristic 3.
- (a) Show that the action of the k -linear Frobenius morphism f on $H^1(C, \mathcal{O}_C)$ is identically 0.
 - (b) Fix a point $P \in C$, and show that there is a nonzero $\xi \in H^1(C, \mathcal{O}_C(-P))$ such that $f^*\xi = 0$ in $H^1(C, \mathcal{O}_C(-3P))$.
 - (c) Now let $\mathcal{E}$ be defined by ξ as an extension

$$0 \longrightarrow \mathcal{O}_C \longrightarrow \mathcal{E} \longrightarrow \mathcal{O}_C(P) \longrightarrow 0,$$

and let X be the corresponding ruled surface over C . Show that X contains a nonsingular curve $Y \equiv 3C_0 - 3f$, such that $\pi : Y \rightarrow C$ is purely inseparable. Show that the divisor $D = 2C_0$ satisfies the hypothesis of (2.21b), but is not ample.

16. Let C be a nonsingular affine curve. Show that two locally free sheaves $\mathcal{E}, \mathcal{E}'$ of the same rank are isomorphic if and only if their classes in the Grothendieck group $K(X)$ are the same. This is false for a projective curve.

Proof. Let A be a Dedekind domain that is finitely generated over k such that $C \cong \text{Spec } A$. The category of coherent $\mathcal{O}_C$ -modules is equivalent to the category of finitely generated A -modules. Under this identification, $K(X)$ is naturally isomorphic to the Grothendieck group of flat finitely generated A -modules $K_0(A)$ (A.M., Ex. 7.26). The structure theorem of modules over Dedekind domains states $K_0(A) \cong \mathbb{Z} \oplus \text{Cl}(A)$, where if M is any flat finitely generated A -module, then $M \cong A^r \oplus I$, where K is the fraction field of A , $r + 1 = \dim_K M \otimes_A K$ is the rank of M , and I is some fractional ideal in $\text{Cl}(A)$. If $N \cong A^s \oplus J$ is another A -module, then $M \cong N$ if and only if $r = s$ and $I \cong J$ in $\text{Cl}(A)$. $\square$

17. (a) Let $\varphi : \mathbb{P}_k^1 \rightarrow \mathbb{P}_k^3$ be the 3-uple embedding. Let $\mathcal{I}$ be the sheaf of ideals of the twisted cubic curve C which is the image of φ . Then $\mathcal{I}/\mathcal{I}^2$ is a locally free sheaf of rank 2 on C , so $\varphi^*(\mathcal{I}/\mathcal{I}^2)$ is a locally free sheaf of rank 2 on $\mathbb{P}^1$. By (2.14), therefore, $\varphi^*(\mathcal{I}/\mathcal{I}^2) \cong \mathcal{O}(l) \oplus \mathcal{O}(m)$ for some $l, m \in \mathbb{Z}$. Determine l and m .
- (b) Repeat part (a) for the embedding $\varphi : \mathbb{P}^1 \rightarrow \mathbb{P}^3$ given by $x_0 = t^4, x_1 = t^3u, x_2 = tu^3, x_3 = u^4$, whose image is a nonsingular rational quartic curve.